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# S1 Continuous-Duty Motor Thermal Sizing Worksheet

For OEM designers and plant engineers specifying PMDC or BLDC motors for 24/7 operation

Work through the six steps in order for each continuous-duty drive. Use the manufacturer's test data (torque constant, resistance at 20 °C, thermal resistance, no-load losses) wherever it is available, and keep the completed sheet with the machine's technical file.

## Step 1 — Define the duty and the environment

- [ ] Confirm the duty type: any load that runs longer than about three thermal time constants is S1 continuous duty.
- [ ] Record running hours per day and days per year (24/7 operation is 8,760 h per year).
- [ ] Measure the ambient at the motor position, inside any enclosure, during the hottest month; flag anything above 40 °C.
- [ ] Record the site altitude and flag any installation above 1,000 m.
- [ ] Note dust, moisture, washdown or monsoon exposure for the IP code selection.
- [ ] Record the nominal supply voltage and its lowest value under load.

## Step 2 — Calculate RMS torque at the motor shaft

- [ ] List every segment of the load cycle: torque Tk (Nm) and duration tk (s), including acceleration torque J × alpha.
- [ ] Refer load torque to the motor shaft: divide output torque by (gear ratio × gearbox efficiency).
- [ ] T_rms = square root of [sum of (Tk² × tk) ÷ sum of tk].
- [ ] Also calculate the average torque; never size the motor on the average.
- [ ] Check the peak torque against the motor's intermittent zone and the controller current limit.
- [ ] Convert to current: I_rms = T_rms ÷ Kt, using Kt in Nm/A from the test sheet.

## Step 3 — Build the loss budget

- [ ] Copper loss = I_rms² × R_hot, where R_hot = R20 × [1 + 0.00393 × (T − 20)].
- [ ] Start from an assumed winding temperature of 120 °C and iterate once the rise is known.
- [ ] Brush loss (PMDC only) ≈ 2 V × average current.
- [ ] Iron, friction and windage loss: take from the no-load test at operating speed.
- [ ] PMDC: allow for ferrite flux loss of about 0.18–0.2 % per °C of magnet temperature rise (more current for the same torque).
- [ ] Add gearbox losses if the gearbox is flange-mounted to the motor.

## Step 4 — Predict temperature rise and apply derating

- [ ] Temperature rise (K) = total loss (W) × thermal resistance R_th (K/W).
- [ ] Take R_th from the supplier's heat-run data, not from a catalogue estimate.
- [ ] Repeat the calculation at the new winding temperature until two passes agree within 2 K.
- [ ] Design target: rise ≤ 80 K at rated ambient on Class F insulation (Class B rise).
- [ ] Add a 10 K hot-spot allowance and check against 130 °C (B), 155 °C (F) or 180 °C (H).

- [ ] For ambient above 40 °C, reduce the permissible rise by 1 K for every °C of excess.
- [ ] Above 1,000 m, reduce the permissible rise by about 1 % per 100 m unless the supplier states otherwise.

## Step 5 — Select the cooling method and enclosure

- [ ] IC410 (TENV): speed-independent cooling; suits variable-speed drives; lower output per frame.
- [ ] IC411 (TEFC): airflow falls roughly in proportion to speed; derate for constant torque at low speed.
- [ ] IC416 (independent fan): full cooling at any speed, including torque at standstill.
- [ ] Specify the IP code and IC code together; tighter sealing usually lowers the continuous rating.
- [ ] On small frames, use B5 or B14 flange mounting onto a metal structure as an extra conduction path.
- [ ] Keep fans and fins clear of guards and adjacent heat sources; avoid recirculating hot exhaust air.

## Step 6 — Protect and verify

- [ ] BLDC: embed PTC or KTY/PT1000 sensors in the stator end windings.
- [ ] PMDC: use a frame or brush-gear sensor plus a controller $I^2t$ current model.
- [ ] Avoid auto-reset bimetal protectors wherever an unexpected restart would be unsafe.
- [ ] Run a heat run on the prototype at the worst-case load cycle until temperature changes by less than 2 K per hour.
- [ ] Measure rise by resistance: rise = (R_hot − R_cold) ÷ R_cold × (235 + T_cold) + T_cold − T_ambient.
- [ ] Record the hot-spot margin and file the result with the machine's technical documentation.

## Indicative continuous-torque derating (motor designed to an 80 K rise, about 30 % fixed losses)

| Ambient at motor | Rise allowance | Torque factor | Example: 3.5 Nm rating |
|---|---|---|---|
| 40 °C | 80 K | 1.00 | 3.50 Nm |
| 45 °C | 75 K | 0.95 | 3.34 Nm |
| 50 °C | 70 K | 0.91 | 3.17 Nm |
| 55 °C | 65 K | 0.86 | 3.00 Nm |
| 60 °C | 60 K | 0.80 | 2.81 Nm |

*Values are indicative engineering estimates. Confirm torque constant, resistance, thermal resistance and derating factors against the manufacturer's test data for the specific motor before finalising a design.*

**Need help sizing or replacing a motor?**
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

*General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.*